

Kirloskar's TURBOTEK & PRODIGY Chillers with their advance feature Becomes the obvious choice for 'Data Centre Cooling'

When Kirloskar TURBOTEK Centrifugal or PRODIGY Screw Chillers are used for cooling of Data Centers, certain features are built into the chillers. The basic philosophy is to have minimum stoppage to avoid the rise in temperatures during power shut down. The very reason to maintain temperature in the Data Centers is to ensure error-free, efficient & effective storing, processing & transmission of data.

The features and corresponding elaboration is given below -

- UPS Power Supply
 - When chillers are to be used for data centers, the power supply to the controller is always through a UPS. The UPS is unit mounted and factory wired. This is to eliminate the boot up time required for the controller in case of a power failure.
 - The advanced software logic senses power loss and re-starts the unit on resumption of power supply.
 - Proper start up sequence is initiated on return of power supply, ensuring availability of water flow through evaporator and condenser, start up of oil pumps, etc. This ensures that all the working requirements for a safe start are ensured and at no point of time, the safety of the chiller is jeopardized.
- Refrigerant Pressure Equalisation
 - On power loss, and stoppage of compressor, controller initiates pressure equalisation cycle to equalise evaporator and condenser refrigerant pressures. This is done through opening of the electronic expansion valve, thus increasing the suction pressure.
 - This ensures smooth start up of the chiller, eliminating any nuisance trips due to low suction pressure or avoidance of high starting kick due to compressor starting against high lift.
- Immediate Startup
 - On power resumption, the controller immediately starts the start up sequence.
 - Proper start up sequence is followed and within 40 sec from resumption of power, compressor motor starts. During the start up sequence, the oil pump(for centrifugal chillers) starts first, and along with it the compressor unloading also starts. After further 15 seconds, the standard start up logic starts, checking the flows, oil pressures, etc and if all the safeties are healthy, the compressor is powered on at the end of the 40th second.

Kirloskar Chillers Private Limited

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- Meanwhile, the controller is also keeping a count of total starts in an hour and if these exceed three starts, then controller shall not initiate any other start in that hour to protect the compressor motor.

- Fast Loading

- At the time of the power loss, the controller also records and keeps in its memory the load on each compressor (for multi compressor units) and percent opening of the electronic expansion valves. After start up, the compressor is loaded to this pre-stoppage load level under fast load sequence, and then depending on leaving chilled water temp, further loading and unloading is initiated.
- The fast load sequence can load the compressor from start up to 100% level within two minutes.
- Thus the complete start up sequence is completed with in 180 seconds from the resumption of power.

Below figure (Fig.1) makes the sequence more clear in terms of time line of the entire start up process, in case of power shut down.

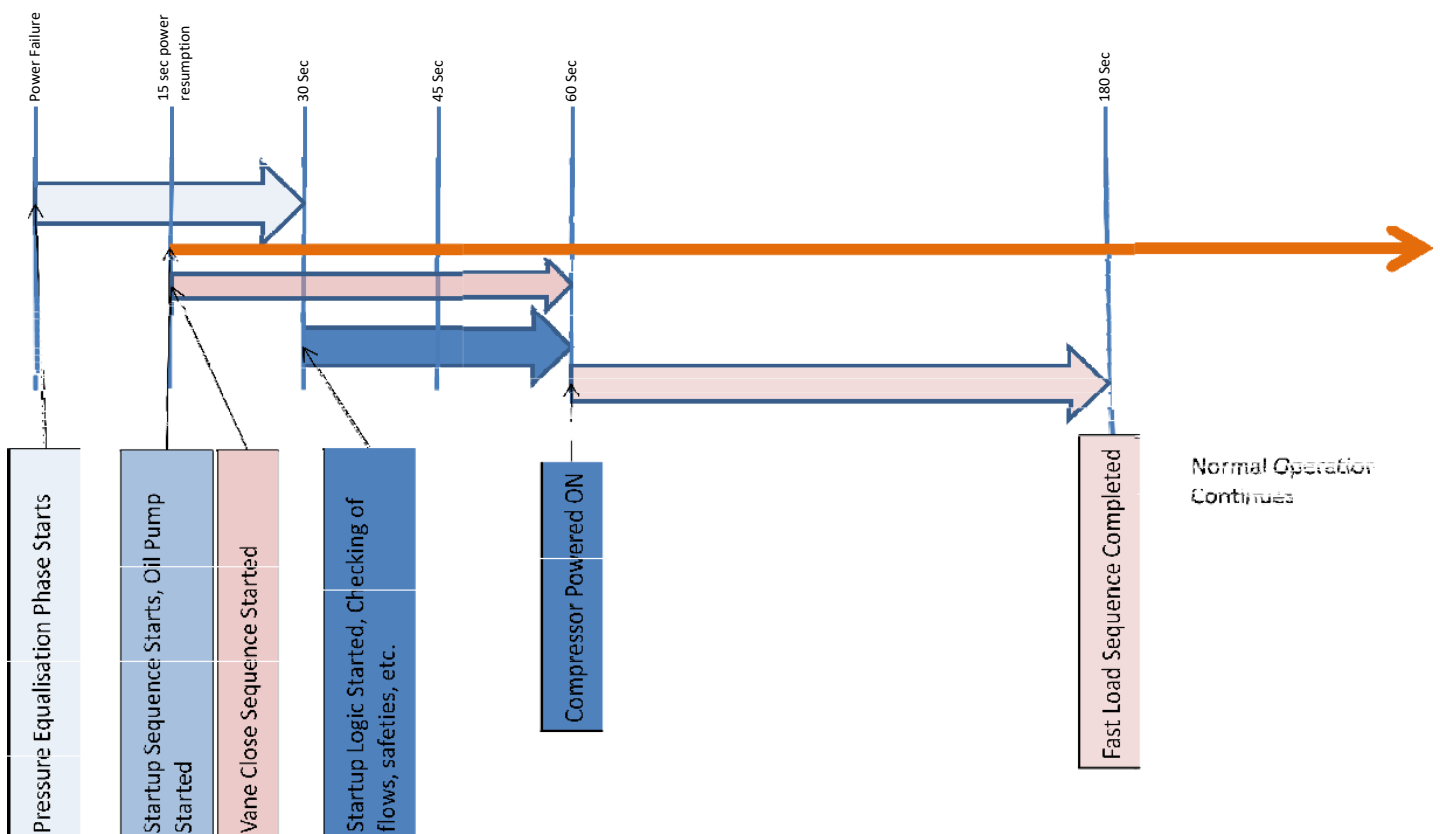


Fig. 1

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